

M003 Week 4 — English Worksheet (Beginner)

Topic: The Ground Plane (x, z) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

For three weeks you drew standing up on a **wall** with (x, y). This week you look **down** at the **floor**. On the floor the two numbers are **(x, z)**: x is still how far **across**, and the new number **z** is how far **forward**.

1 · Predict

Read the steps. Before you imagine it happening, circle or write your answer.

```
place gold block at (3, 0)
place gold block at (3, 5)
```

On the floor these are (x, z). Both have the same x. Do they land in the same spot? Circle one:
yes · no

Why?


```
place red block at (0, 0)
```

Counting starts at 0. Is (0, 0) the corner where the floor begins? Circle one: yes · no

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The floor block should land at **(2, 5)** — 2 across on x, 5 forward on z.

```
# clean
place gold block at (2, 5)
```

```
# buggy
place gold block at (5, 2)
```

What is wrong? Which two numbers got mixed up?

Pair B — The path should go **forward**, so the **z** number must grow.

```
# clean
set z to 0
repeat 4 times:
  place block at (3, z)
  add 1 to z
```

```
# buggy
set x to 0
repeat 4 times:
  place block at (x, 0)
  add 1 to x
```

What is wrong? Which way does the buggy path go — forward or across?

3 · Fill the Gap

The path should go **straight forward** at $x = 3$. The **z** number must grow each time. One word is missing. Fill it in using the word bank.

```
set z to 0
repeat 4 times:
  place block at (3, z)
  add 1 to ____
```

Word bank: z · x · block

Write the missing word:

4 · Tell Me What You Built 📷

Now switch to your homework world. On the floor, build a **path** going forward from (0, 0), then add a second path going **across** so they make a plus or a corner. When you finish, come back here.

Send a photo or video of your floor, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My forward path grew the number ...

My across path grew the number ...

The two paths meet at ...

5 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while you look down at your floor. Talk like you are teaching a friend. Try to use these words: **floor**, **x**, **z**, **forward**, **across**.

1. Show the block at (0, 0) where the floor begins.
2. Walk along your forward path and say which number is growing.
3. Say in your own words what **x** and **z** mean on the floor.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.